

Siamese Transformer Networks for Edit Distance Embedding and Evaluation

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Abstract

This report evaluates the efficacy of a Siamese Transformer architecture in mapping error correction code operations into a geometric embedding space. Initial experiments demonstrated that standard generative decoding struggles with edit distance tasks due to shift-induced error propagation. By utilizing an encoder-only architecture trained with Triplet Margin Loss, the network successfully learns the structural relationships between sequences without autoregressive decoding. The model achieves a 0.7238 Triplet Accuracy, demonstrating that attention mechanisms can effectively map discrete sequence mutations into a measurable metric space, outperforming standard recurrent and convolutional baselines.

Methodology and Architecture

Standard generative sequence-to-sequence models face fundamental limitations when applied to edit distance problems. A single uncorrected insertion or deletion operation shifts the entire predicted sequence, causing standard positional loss functions to penalize structurally valid outputs. To resolve this instability, the architecture was refactored into an encoder-only Siamese network designed to evaluate Euclidean distances in a continuous manifold.

The dataset utilizes a triplet schema consisting of anchor, positive (correlated), and negative (corrupted) sequences. The core architecture is a Siamese Transformer featuring an embedding dimension of $d_{model} = 256$, utilizing 8 attention heads across 4 encoder layers. Mean pooling is applied across the sequence dimension to collapse the high-dimensional encoder memory into a single representative embedding vector.

The network was trained over 30 epochs using the AdamW optimizer. Optimization was governed by a Triplet Margin Loss function, defined as:

$$\mathcal{L}(a, p, n) = \max\{d(a_i, p_i) - d(a_i, n_i) + \text{margin}, 0\} \quad (1)$$

where d is the pairwise Euclidean distance, a_i is the anchor, p_i is the positive sample, n_i is the negative sample, and the margin is set to 1.0. This function iteratively forces the network to minimize the distance between related sequences while maximizing the distance between unrelated sequences.

Empirical Results

Performance was evaluated by calculating the Euclidean pairwise distances for 10,000 sequence triplets. The network yielded an average positive distance of 2.7128 and an average negative distance of 3.3750, resulting in a Triplet Accuracy of 0.7238.



Figure 1: Distribution of Euclidean distances in the 256-dimensional embedding space, demonstrating clear structural separation between correlated and corrupted sequence pairs.

Figure 1 visualizes the efficacy of the Triplet Margin Loss. The mathematical separation between the mean distributions confirms the network successfully pulled correlated sequences together while pushing corrupted sequences apart.



Figure 2: Correlation mapping between true Levenshtein edit distance and the predicted Euclidean embedding distance.

Figure 2 demonstrates a positive correlation between the true edit distance and the learned metric space. As the physical corruption severity increases, the Euclidean distance in the embedding space scales proportionally.



Figure 3: Principal Component Analysis projecting the 256-dimensional Siamese embedding manifold into a 2D space, colored by sequence corruption severity.

The Principal Component Analysis in Figure 3 confirms that the attention layers are capturing the underlying variance of the edit distance manifold, organizing sequences spatially based on structural integrity.

Discussion and Baseline Comparison

To contextualize the performance of the Siamese Transformer, the empirical accuracy was benchmarked against established architectural data from the REBIND evaluations. Table 1 outlines the

comparative Triplet Accuracy across the tested models.

Table 1: Comparative Triplet Accuracy Across Network Architectures

Model	Triplet Accuracy
Transformer (Generative Baseline)	0.530
GRU	0.636
CNN-ED	0.684
CGK	0.721
Siamese Transformer (Proposed)	0.7238
REBIND	0.835

The data indicates that standard sequence models struggle to parse the edit distance manifold. A traditional generative Transformer baseline acts almost randomly with an accuracy of 0.530. Recurrent and convolutional architectures achieve 0.636 and 0.684, respectively.

By applying triplet loss to the pooled attention states, the proposed Siamese architecture completely bypasses the limitations of autoregressive decoding. The 0.7238 accuracy significantly outperforms the CNN and GRU models and slightly exceeds the CGK baseline. While the highly specialized REBIND architecture maintains the highest fidelity at 0.835, the variance captured by the Siamese encoder proves that multi-head attention mechanisms can effectively learn the fundamental geometric rules of error correction coding.